Jad El Badaoui

Aerospace Engineer | Structural Analysis, FEA and Composite Structures

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Profile

First Class BEng Aerospace Engineering graduate from the University of Bristol, top of the cohort in Structures and Materials (94%) and Engineering Mathematics (94%), progressing to Imperial College London's MSc in Advanced Computational Methods for Aeronautics. Structural work spans composite testing, Abaqus finite element analysis, cohesive-zone fracture modelling and aerodynamic load transfer, including roughly doubling Mode II delamination resistance at the Bristol Composites Institute and predicting interlaminar failure within 10% of measurement.

Education

Imperial College London London, United Kingdom | Oct 2026 – Oct 2027

MSc Advanced Computational Methods for Aeronautics, Flow Management and Fluid-Structure Interaction

- Incoming student, October 2026. Core computational mechanics and fluid-structure interaction, computational linear algebra and numerical methods; options in finite elements, aeroelasticity and high-performance computing.

University of Bristol Bristol, United Kingdom | Sep 2023 – Jul 2026

BEng Aerospace Engineering, First Class Honours

- Highest mark in the cohort in **Structures and Materials (94%)**, Engineering Mathematics (94%) and Aerodynamics (95%).
- Specialised in advanced structures, mechanics of materials and composites. Final-year research project in composite laminate strength supervised by Prof. Michael R. Wisnom.

Technical Skills

Structural Analysis and FEA: Abaqus/CAE (3D ply-by-ply models, cohesive-zone elements, free-edge stress analysis), ANSYS Mechanical; static stress from first principles to FEM-based analysis, reserve factors at limit and ultimate load, orthotropic composite modelling, aerodynamic load transfer, verification and validation.

Composite Mechanics and Materials: Classical laminate theory and ABD matrices, Hashin failure criteria, O'Brien strain energy release rates, delamination and interlaminar failure, through-thickness reinforcement, thin-ply and Double-Double architectures.

Structural Testing: Instron load frames, video-gauge and extensometer strain measurement, Mode II delamination testing, static test to ultimate load, SEM fractography, statistical treatment of test scatter, test-to-model correlation.

Composite Manufacture: Spread-tow prepreg layup, vacuum bagging, autoclave cure, pultrusion-assisted hand layup, wet and dry fibre layup over tooling, FDM additive manufacture of flight parts.

Programming and CAD: Python, MATLAB, C++; data reduction pipelines, automated testing of numerical code; Autodesk Inventor, SpaceClaim.

Structural Research and Engineering Experience

Bristol Composites Institute, University of Bristol Bristol, United Kingdom | Jun 2025 – Aug 2025

Research Intern, Through-Thickness Reinforcement of Composites

- Designed interlaminar mechanical interlocks to raise delamination resistance and damage tolerance under transverse loading, selecting concepts for prototyping from a review of the through-thickness reinforcement literature.
- Manufactured the prototypes by pultrusion-assisted hand layup and autoclave cure, so competing reinforcement concepts could be judged on test data early in the design cycle.
- Planned and ran the test programme and its analysis with custom Python data reduction, **roughly doubling Mode II delamination resistance** with the new architecture.
- Modelled the laminates in Abaqus using competing approaches, including nonlinear cohesive-zone fracture analysis, **predicting interlaminar failure within about 10% of measurement**.

Thin-Ply Quad and Double-Double Composite Laminates — 2025 – 2026

Final-year research project (AENG30017), supervised by Prof. Michael R. Wisnom

- Established whether the unnotched tensile strength penalty of Double-Double laminates survives ply thinning, comparing blocked against dispersed stacking sequences across four stiffness-equivalent 32-ply architectures.
- Manufactured and autoclave-cured the laminates from TC33/K51 spread-tow prepreg with 0.03 mm plies, then tested them to failure on an Instron 1342 with video-gauge strain measurement: 496 to 538 MPa at 1.41 to 1.53% strain.
- Built a ply-by-ply 3D Abaqus free-edge model alongside classical laminate theory, computing O'Brien strain energy release rates of 45 to 254 J/m^2 against a 200 to 500 J/m^2 critical range, and identified the failure mechanisms by SEM fractography.
- Ran the statistical analysis establishing whether the differences between architectures are real, and backed the pipeline with 65 automated tests.

HyPower Bristol — Bristol, United Kingdom | Oct 2024 – Sep 2025

Aerodynamics and Structures Engineer (part-time)

- Reduced the worst-case flight condition to an equivalent **146 kPa** aerodynamic pressure and applied it to an Abaqus model of the three-ply triaxial [-45/0/45] carbon fibre fins, checking the laminate under flight loading rather than a nominal case.
- Manufactured fins and a nose cone in carbon fibre by dry fibre and wet layup over 3D printed moulds, iterating tooling and process until surface finish and fitment were flight acceptable.
- Built the full-scale STAR-CCM+ models that generated those loads, running a 25-case sweep of airbrake deployment against velocity and verifying it against closed-form boundary layer theory to within 13%.

Selected Engineering Projects

Blood-Transport UAV Airframe (AVDASI 2) — 2024 – 2025

Structural Joints and Fairings Lead, and Project Manager, 60-student programme

- Delivered every structural joint and aerodynamic fairing on a full-scale 2.6 m UAV: ten FDM-printed flight parts at 1,011 g total, integrated inside a 535-file Autodesk Inventor vehicle assembly under configuration control.
- Computed **reserve factors for the** $6g \times 1.5$ **ultimate load case, governing value 1.95**, and substantiated the joint sizing against the requirement set.
- Static-tested the airframe past ultimate to 813 N, where it yielded in a printed part the analysis had treated as a non-structural former; reported the scoping failure and fed it back into the model.

Composite Rotor Blade under Aerodynamic and Centrifugal Loading — 2026

Independent study, one-way fluid-structure interaction in ANSYS 2026 R1

- Mapped a pressure field and centrifugal load onto an orthotropic composite shell blade in ANSYS Mechanical as a one-way coupling, solving the rotor in a rotating reference frame on a 120° periodic sector: 0.405 m tip deflection.
- Verified the root radial reaction to within 0.116% of the analytic $m\omega^2 r$, and validated the companion aerofoil solution against NASA measurement to within 1.4% using Richardson extrapolation and the grid convergence index.

Saad & Trad S.A.L. and M. Ezzat Jallad & Fils S.A.L. — Lebanon | Jun 2024 – Jul 2024

Engineering Intern

- Two one-month workshop placements: assembly, disassembly and preventive maintenance on high-performance vehicles, then maintenance, engine strip and repair of Caterpillar heavy machinery and diesel generator sets.

Awards and Activities

- **Winner (Local) and Global Nominee, NASA International Space Apps Challenge**, 2025. Satellite-data land use suitability tool for Addis Ababa planners.
- **3rd Place, National Rocketry Championship**, 2024. Co-led the team that designed, built and launched a rocket carrying a Geiger counter payload measuring radiation with altitude.
- **Assembling and Testing a Quadcopter**, MathWorks verified certificate, 2024. Subsystem integration, sensors and data acquisition in MATLAB and Simulink.
- **Team Lead, Engineers Without Borders**, 2023. Water supply and purification scheme for Pu Ngaol, Cambodia, down-selected on maintainability. Event Organiser, Lebanese Society, 2023 to 2026; AeroSoc member throughout the BEng.